DNA DAMAGE AND REPAIR

More than 100 DNA repair genes have been identified. The **nucleotide excision repair (NER)** pathway acts on DNA damaged by UV radiation, repairing **cyclobutane pyrimidine dimers (CPDs)** and other photoproducts, as well as DNA damaged by certain carcinogens such as benzo-[a]-pyrene. In NER, the damaged segment of DNA is removed and replaced with undamaged DNA through the coordinated action of multiple proteins (see Fig. 109-3). Defects in these repair genes can lead to human diseases, including **xeroderma pigmentosum (XP)**, **Cockayne syndrome**, and **trichothiodystrophy** (see Chap. 140 for details).

For example, XP can result from mutations in any one of several genes involved in the NER pathway. Complementation groups are defined by cell fusion experiments: if fusion of two cells restores normal DNA repair, the cells are said to "complement" each other and belong to different complementation groups. Seven such groups have been identified (XPA to XPG), indicating that mutations in any of seven distinct genes can cause XP.

Transcribed regions of the genome are repaired more rapidly than nontranscribed regions. This difference arises because the initial steps of damage recognition in NER differ between transcribed and nontranscribed DNA:

- In **global genome NER (GG-NER)**, which operates throughout the genome, the products of the **XPC** and **XPE** genes recognize and bind to UV-damaged DNA, marking it for repair.
- In **transcription-coupled NER (TC-NER)**, damage is detected when RNA polymerase stalls at a lesion in a transcribed gene. This stall recruits specific repair factors to initiate repair preferentially in actively transcribed genes.

This preferential repair of transcribed genes helps preserve essential genetic information and maintain cellular function.

Defense Mechanisms Against UV-Induced Damage

In addition to DNA repair, several intrinsic defense mechanisms protect against the carcinogenic effects of UV radiation:

- **Cell cycle arrest**: DNA damage triggers a temporary halt in the cell cycle, allowing time for repair before replication.
- **Apoptosis (programmed cell death)**: Cells with severe, irreparable DNA damage undergo apoptosis, eliminating potentially malignant clones.
- **Increased melanogenesis**: UV exposure stimulates melanin production, which absorbs and dissipates UV radiation.
- **Epidermal thickening**: The epidermis and stratum corneum thicken in response to chronic UV exposure, providing a physical barrier.
- **Immune surveillance**: The immune system recognizes and removes mutated or abnormal cells, reducing the risk of tumor development.

These mechanisms collectively reduce the accumulation of UV-signature mutations—such as $\mathbf{C} \rightarrow \mathbf{T}$ and $\mathbf{CC} \rightarrow \mathbf{TT}$ base substitutions—and lower the risk of skin cancer development following UV exposure.